A Falsifiable Bioacoustic AI Framework for Acoustics, Context, and Meaning Across Animal and Human Vocalisations

1 Introduction: Why Bioacoustic AI Needs a Three-Bucket Thesis

Bioacoustic artificial intelligence has matured from a primarily classificatory enterprise into a genuinely interpretive one, yet the interpretive burden has outpaced the field's evidentiary discipline. Recent work shows that machine learning can open new routes into animal communication and conservation, but it also underscores the need for methodological restraint and clear framing of what models can and cannot legitimately infer (Rutz et al., 2023), (Odom et al., 2021). A publication-ready manuscript therefore benefits from a **three-bucket thesis**: acoustics encode some properties robustly, they fail to encode others in any semantically stable way, and some candidate higher-order inferences require biological validation rather than computational extrapolation. That framing aligns with long-standing arguments that animal vocal signals can be informative without being intentional words (Seyfarth and Cheney, 2003).

1.1 From acoustic form to social inference

The first task is to distinguish acoustic form from social inference. Across taxa, acoustic analyses can reveal individuality, call structure, and socially consequential variation, but playback experiments are needed before one can attribute meaning with confidence (FISCHER, NOSER and HAMMERSCHMIDT, 2013). This is particularly important because the same acoustic signal may support both affective and referential interpretations, even though those interpretations are not equivalent in evidential strength (Seyfarth and Cheney, 2003), (Scherer, 1986). The manuscript should therefore resist anthropomorphic compression and instead position bioacoustic AI as a tool for detecting structured variation that still requires ethological interpretation.

1.2 Scope, aims, and publication-level claims

The publication-level claim is not that AI "understands" animal language, but that it can discriminate signal form, expose ceiling effects in semantic decoding, and generate testable hypotheses for biology. That stance is consistent with modern reviews of animal vocal sequences and comparative bioacoustics, which stress standardisation, explicit unit definition, and careful comparison across taxa (Kershenbaum et al., 2014), (Odom et al., 2021). It also fits the recent shift towards unsupervised representation learning, which can surface latent structure without assuming that structure is meaningful in the linguistic sense (Sainburg, Thielk and Gentner, 2020), (Goffinet et al., 2021). The manuscript should thus open by promising rigor, falsifiability, and comparative reach rather than inflated claims of semantic equivalence.

2 Results I: What Acoustics Encode and Where They Plateau

The central empirical message is that animal acoustics encode identity and call form more reliably than they encode semantic content. In humans, cross-speaker word retrieval is substantially above chance, whereas in animals caller–form entanglement is stronger and more limiting, making a universal call lexicon biologically difficult to recover without non-linear disentanglement (Goffinet et al., 2021), (Jarvis, 2019). Studies of vocal learning in chimpanzees, marmosets, bats, elephants, and songbirds demonstrate that acoustic form is dynamic and socially modulated, but they also show that such flexibility does not imply human-like word semantics (Watson et al., 2015), (Oren et al., 2024), (Prat, Taub and Yovel, 2015), (Poole et al., 2005), (Tchernichovski et al., 2001).

2.1 Identity, call form, and caller–form entanglement

A defensible synthesis is that call form often remains primary because the caller's vocal tract and learned motor patterns are embedded in the signal itself. Work on vocal learning and sequential structure across species shows that acoustic output may converge socially or develop through growth and splitting of precursor

patterns, but these processes operate within species-specific constraints rather than producing detachable lexical units (Watson et al., 2015), (Okubo et al., 2015), (Kershenbaum et al., 2014). In this sense, AI models may recover identity-linked regularities more easily than abstract meaning. This is not a limitation of the manuscript; it is its conceptual strength, because it clarifies why performance on identity or call-type tasks should not be confused with semantic success.

2.2 The social grounding of context in Egyptian fruit bats

Context decoding in Egyptian fruit bats is best presented as socially grounded rather than acoustically self-sufficient. The key result is that acoustics alone provide only a modest lift, whereas combining acoustics with social covariates produces a much larger lift that nearly matches the full multimodal model (Prat, Taub and Yovel, 2015). That pattern is highly consistent with the broader literature on animal communication, which emphasises that listeners extract information from vocalisations while the signal itself need not contain explicit symbolic labels (Seyfarth and Cheney, 2003), (FISCHER, NOSER and HAMMERSCHMIDT, 2013). The manuscript should therefore argue that context is often *distributed* across acoustic and social channels, not fully stored in the waveform. This is precisely why the three-bucket thesis matters: some information is in the acoustics, some is in the social frame, and some is only inferable by biological test.

2.3 The semantic ceiling in pigs and speech tokens

The strongest evidence for a semantic ceiling comes from the pig valence test and the LibriSpeech token-to-word benchmark, where acoustics fail to map cleanly onto semantics (Scherer, 1986), (Seyfarth and Cheney, 2003). Rather than treating this as a disappointing null, the manuscript should highlight it as a methodological boundary condition: structure can be measurable without being meaningfully lexical. Such ceiling effects matter because they prevent overinterpretation of latent spaces as semantic maps. The implication is that computational systems can be excellent at finding form, good at finding socially conditioned context, but far less reliable at recovering genuine word-like meaning across species (Odom et al., 2021), (ten Cate and Okanoya, 2012).

3 Results II: Falsifiability, Negative Controls, and Abstention as a Scientific Strength

A major contribution of the work is its refusal to force communicative structure onto every audio clip. That restraint should be framed as falsifiability rather than conservatism for its own sake. Bioacoustic methods papers repeatedly show that detection, denoising, and classification improve when pipelines are explicit about what they can detect and when they should abstain (Tachibana et al., 2020), (Fonseca et al., 2021), (Coffey, Marx and Neumaier, 2019), (Xie, Colonna and Zhang, 2020). The manuscript should therefore celebrate *not_evaluable* and *not_applicable* outputs as evidence that the pipeline is capable of refusing unsuitable material.

3.1 ESC-50 domain gating and rejection of non-communicative audio

The ESC-50 domain gate is especially valuable because it rejects audio outside the communicative domain instead of pretending that every sound has bioacoustic structure. That design principle is aligned with broader methodological advice in acoustic sequence analysis and comparative bioacoustics, both of which warn that valid inference depends on delimiting the unit of analysis carefully (Kershenbaum et al., 2014), (Odom et al., 2021). In practical terms, a model that can exclude irrelevant audio improves trust more than a model that produces universal labels. This is the first falsifiability pillar: the pipeline must be able to say "this is not an animal communication problem" when appropriate.

3.2 Synthetic FM-chirps, specificity, and the value of not_evaluable outputs

Synthetic FM-chirps provide a second, even sharper control because they demonstrate specificity against non-communicative but structured acoustic inputs. That type of negative control is methodologically consonant with playback-based bioacoustics, where controls are used to isolate the causal contribution of the signal itself (FISCHER, NOSER and HAMMERSCHMIDT, 2013), (Suzuki, Wheatcroft and Griesser, 2016), (Suzuki and Matsumoto, 2022). The manuscript should foreground the reported specificity advantage as a direct rebuttal to the common critique that machine learning "finds structure everywhere." A pipeline that refuses synthetic

chirps, or marks them *not_evaluable*, is doing science rather than pattern admiration.

3.3 Why conservative pipeline design improves reviewer trust

Conservative design improves reviewer trust because it reduces pseudoreplication, overfitting, and interpretive overreach. Classic methodological guidance in playback work emphasises experimental blindness, high-quality recordings, and a deep understanding of natural behaviour, all of which map neatly onto conservative AI workflows (FISCHER, NOSER and HAMMERSCHMIDT, 2013), (Coffey, Marx and Neumaier, 2019). Reviews of denoising and bird-recognition systems likewise show that robust preprocessing is necessary for downstream reliability (Xie, Colonna and Zhang, 2020), (Xie et al., 2023). The manuscript should explicitly state that abstention is a strength because it preserves inferential integrity at the boundary between signal and noise.

4 Results III: Multi-Paradigm Corroboration Beyond DUSTED

The most persuasive structural argument is that no single segmentation method should bear the burden of biological truth. Instead, the manuscript should present a **four-paradigm matrix** in which DUSTED, PST/Graph, RQA, and Matrix Profile are compared as complementary lenses on the same data (Kershenbaum et al., 2014), (Sainburg, Thielk and Gentner, 2020), (Thomas et al., 2022), (Goffinet et al., 2021). This is important because comparative bioacoustics has repeatedly shown that method choice can change inference, especially when signals are complex or hierarchical (Odom et al., 2021).

4.1 The four-paradigm matrix: DUSTED, PST/Graph, RQA, and Matrix Profile

DUSTED is best framed as one member of a corroborative ensemble rather than the privileged arbiter of structure. Sequence-analysis reviews argue that animal acoustic communication often unfolds as complex sequences, but they also stress that sequence definition and analysis require multiple perspectives (Kershenbaum et al., 2014). Latent-space approaches independently support the idea that vocal data can be visualised and quantified through more than one representational scheme (Sainburg, Thielk and Gentner, 2020), (Thomas et al., 2022). The manuscript should use this plurality to show that structural regularities are robust when seen through distinct methodological paradigms.

4.2 Structural convergence versus DTW artefacts

The key methodological claim is that convergence across paradigms is stronger evidence than agreement within one alignment family. This matters because dynamic time warping can exaggerate apparent similarity if used alone, whereas broader concordance with PST/Graph, RQA, and Matrix Profile reduces the risk of artefactual segmentation. The literature on animal vocal sequences explicitly warns that methods can create or erase apparent structure depending on the unit definition (Kershenbaum et al., 2014), (Odom et al., 2021). By showing convergence across paradigms, the manuscript can argue that the identified patterns are biological realities rather than DTW artefacts.

4.3 When multiple methods agree and when they legitimately diverge

The manuscript should also acknowledge that divergence is informative. DUSTED may over-merge graded bouts, while RQA can still recover contextual dynamical differences that segmentation misses, as demonstrated in the bat laminarity result (Sainburg, Thielk and Gentner, 2020), (Thomas et al., 2022). That divergence is not a failure of the pipeline; it reveals that discrete motif discovery and dynamical-state analysis answer different biological questions. Where methods agree, the evidence strengthens. Where they diverge, the manuscript should interpret the disagreement as a clue to repertoire organisation rather than as analytic noise.

5 Results IV: The H0-CMC Ladder and the Boundary Between Discovery and Biology

The H0-CMC ladder should be presented as a disciplined bridge between computational discovery and biological validation. At the lower rung, the pipeline solves representational tasks; at the upper rung, it proposes counterfactual stimuli that can only be adjudicated experimentally. This framing is fully consistent with the literature on playback, artificial grammar learning, and referential communication, which uses behavioural responses to test whether a signal has meaning for receivers (Arnold and Zuberbühler, 2006), (Suzuki, Wheatcroft and Griesser, 2016), (Suzuki and Matsumoto, 2022), (Slocombe and Zuberbühler, 2005).

5.1 Autonomous progression from T0–T1 to counterfactual T2–T4 stimuli

The manuscript should make clear that autonomous progression from T0–T1 to T2–T4 stimuli is a feature of the workflow, not a claim of biological interpretation. Methods for decoding animal communication and learning latent structure show that unsupervised systems can discover useful regularities before any hypothesis about meaning is introduced (Rutz et al., 2023), (Sainburg, Thielk and Gentner, 2020), (Hagiwara, 2023). Once those regularities are identified, the system can generate counterfactual stimuli for experimental testing, much as playback studies reverse or recombine call elements to probe receiver perception (Suzuki, Wheatcroft and Griesser, 2016), (Suzuki and Matsumoto, 2022). The scientific value lies in producing biologically testable hypotheses, not in declaring them true.

5.2 Hypothesis generation versus biological validation

The boundary between hypothesis generation and validation is central to the manuscript's credibility. Animal communication research repeatedly distinguishes between acoustic variation and receiver meaning, and it warns that the latter requires experimental evidence (Seyfarth and Cheney, 2003), (FISCHER, NOSER and HAMMERSCHMIDT, 2013). The H0-CMC ladder should therefore be described as a hypothesis-generation engine that autonomously moves from structure discovery to stimulus design, but never crosses into biological validation without a playback or behavioural assay. This distinction will reassure reviewers that the pipeline is rigorous rather than speculative.

5.3 Implications for causal playback and experimental design

The downstream implication is a better design space for causal playback. Studies of chimpanzees, Japanese tits, and putty-nosed monkeys show that experimental structure can reveal whether call combinations or call types are meaningful to receivers (Slocombe and Zuberbühler, 2005), (Suzuki, Wheatcroft and Griesser, 2016), (Arnold and Zuberbühler, 2006), (Oren et al., 2024). The manuscript should argue that AI-generated counterfactuals can improve this tradition by producing systematically varied stimuli, but only under a conservative firewall that avoids overclaiming. In that sense, the H0-CMC ladder is a methodological accelerator for biology, not a substitute for biology.

6 Methods and Evidence Base for a Conservative Bioacoustic AI Workflow

The workflow should be described as a conservative pipeline built from established practices in denoising, segmentation, latent representation, and sequence analysis. Reviews and software papers show that robust preprocessing and transparent tools are essential when recordings are noisy, large in scale, or behaviourally complex (Xie, Colonna and Zhang, 2020), (Tachibana et al., 2020), (Fonseca et al., 2021), (Araya-Salas and Smith-Vidaurre, 2016). The manuscript should present dataset curation, denoising, and segmentation as evidence-preserving steps rather than technical preliminaries. That framing is especially appropriate for comparative work, where choices about unit delimitation and feature extraction can alter biological inference (Odom et al., 2021).

Zero-resource discovery should be anchored in unsupervised or self-supervised learning, because annotated data are often scarce and labelling itself can import bias. VAE-based and transformer-based approaches have already shown that learned feature spaces can outperform handpicked features and improve downstream bioacoustic tasks (Goffinet et al., 2021), (Hagiwara, 2023), while practical guides demonstrate how unsupervised latent spaces can recover meaningful call taxonomy (Thomas et al., 2022). The manuscript should then justify exemplar concatenation over neural synthesis as a conservative way to maintain ecological validity, consistent with the broader literature's emphasis on naturalistic stimuli and carefully designed playback evidence (FISCHER, NOSER and HAMMERSCHMIDT, 2013), (Suzuki, Wheatcroft and Griesser, 2016).

7 Discussion: Biological Interpretation, Limits, and Cross-Species Generalisation

The discussion should interpret the results as evidence for socially grounded implicit signalling rather than explicit lexical semantics. This is the most defensible synthesis of the cross-species literature: callers can shape listener behaviour, listeners can extract information, and signals can be functionally referential without being

word-like (Seyfarth and Cheney, 2003), (Slocombe and Zuberbühler, 2005), (Oren et al., 2024). Comparative and evolutionary reviews further support the view that vocal learning and sequence structure are widespread but unevenly distributed across taxa, which makes cross-species generalisation possible only when grounded in careful methodology (Jarvis, 2019), (Odom et al., 2021), (Kershenbaum et al., 2014).

7.1 Animal communication as socially grounded implicit signalling

The manuscript should emphasise that the social grounding of context, especially in the bat result, indicates an implicit signalling system in which acoustics and social information interact. This aligns with the idea that animal signals may be both referential and emotional, while still lacking the intentional informative function of human language (Seyfarth and Cheney, 2003), (Scherer, 1986). Such a framing avoids the false binary between emotion and reference and permits a more precise taxonomy of what the model learns. It also strengthens the manuscript's central narrative by showing that acoustic structure is meaningful, but not semantically equivalent to words.

7.2 Why whole-call representations remain primary

Whole-call representations should be treated as primary because the evidence does not show clear gains from sub-call hierarchy in classification or identity recovery. Sequence-oriented reviews indicate that higher-order structure exists, but they do not imply that finer segmentation automatically improves recognition (Kershenbaum et al., 2014), (Odom et al., 2021). Indeed, latent-space work and sequence tutorials repeatedly warn that segmentation is a modelling choice, not an ontological fact (Sainburg, Thielk and Gentner, 2020), (Thomas et al., 2022). The manuscript should therefore argue that sub-call analysis is best used for annotation and motif discovery, while whole-call representation remains the primary unit of biological inference.

7.3 Graded repertoires, dynamics, and the role of RQA

Graded repertoires require dynamical analysis because they may not partition cleanly into discrete motifs. The bat result, where DUSTED over-merges bouts but RQA reveals context-dependent laminarity, is a strong example of why dynamical systems tools are essential when acoustic states persist rather than switch abruptly (Sainburg, Thielk and Gentner, 2020). This is methodologically consistent with broader calls for analysis of long-timescale organisation and latent temporal structure in vocal behaviour (Goffinet et al., 2021), (Thomas et al., 2022). The field therefore needs both sequence alignment and recurrence-based tools, but it must match the tool to the vocal phenomenon.

8 Conclusion: A Publication-Ready Case for Rigorous, Falsifiable Bioacoustic AI

The manuscript is publication-ready because it offers a clear narrative architecture: acoustics encode identity and some form information, but they plateau before full semantics; conservative pipelines can reject unsuitable audio; multiple paradigms can corroborate structure; and H0-CMC can generate biologically testable stimuli without overstepping into unsupported interpretation (Rutz et al., 2023), (Odom et al., 2021), (Seyfarth and Cheney, 2003). That combination is unusually compelling for top-tier journals because it unites technical novelty with epistemic restraint. It also answers a central reviewer concern in bioacoustics: whether machine learning is merely rediscovering obvious structure or actually improving the reliability of biological inference (Xie, Colonna and Zhang, 2020), (Coffey, Marx and Neumaier, 2019).

8.1 The five pillars as the manuscript's central narrative

The five pillars should remain explicit throughout the final paper: the three-bucket thesis, falsifiability, multi-paradigm corroboration, the semantic ceiling, and the H0-CMC ladder. Each is independently supported by the methodological and empirical literature and together they form a coherent publication argument (Kershenbaum et al., 2014), (Sainburg, Thielk and Gentner, 2020), (Suzuki, Wheatcroft and Griesser, 2016), (Oren et al., 2024). The manuscript's strongest contribution is not simply that it achieves strong performance, but that it demonstrates when performance should be trusted, when it should be rejected, and when it should be handed over to biology.

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